

Exam 1: Search

Name:

CSC 480: Artificial Intelligence, Spring 2026

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Note on Points and Grading

Remember that the grade for your exam will be scored on the coarse 0-4 scale, and so you do not need to score 90+% to earn an A. My goal is for these questions to be challenging in order to really assess your ability to solve hard problems based on the material. Try to write down your thought process for possible partial credit to help me genuinely assess your mastery of the material, and try to do your best without worrying too much about getting stuck on any given problem. Especially as this is the first exam, there will be plenty of wiggle room.

I Intelligent Agents

In this part, consider the following agentic system: *Imagine a country with an (at least seemingly) functional democracy. We can model as an agent an incumbent politician trying to get re-elected by a voting constituency. The agent does this by creating public policy in response to voter demand.*

1. (0.25 points) What are the agent's *perceptions* and *actions*? Does the agent have a measure of *utility/performance*? If so, what is it?

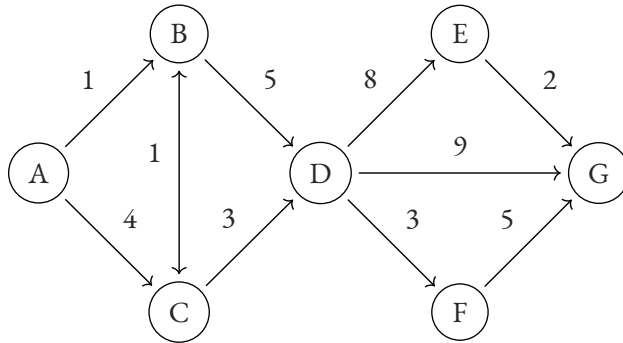
2. (0.25 points) Is the agent best understood as *rational*? If so, in what sense? If not, why not?

II Uninformed Search

3. (0.25 points) In general, under what conditions, if any, is breadth first search an optimal search algorithm?
4. (0.25 points) In a map-coloring problem, the aim is to color in a map of countries using a given set of colors so that no two adjacent countries are the same color. Give a precise formulation of map-coloring as a search problem.
5. (0.25 points) Formulate a precise state space representation for your map coloring search problem. Give a tight upper bound on the size of the state space in terms of the number of countries K and the number of colors C .
6. (0.25 points) Give a tight upper bound on the branching factor of the search problem.

III Informed Search

In this section, consider the state space shown below. Node A is the start state and G is the goal state. Transitions available at each state (note directionality) are shown with associated costs. Two different heuristics, h_1 and h_2 are described with their values at each state, as well as a candidate heuristic h_3 with not all values determined.



Node	h_1	h_2	h_3
A	9.5	10	10
B	9	12	?
C	8	9	9
D	7	8	7
E	1.5	1	1.5
F	4	4.5	4.5
G	0	0	0

7. (0.25 points) Is h_1 admissible and/or consistent? What about h_2 ?

8. (0.25 points) What is the order of search expansions of A^* tree search using heuristic h_2 .

9. (0.25 points) What range of possible values for $h_3(B)$ (i.e. the interval $[0, \infty)$ for all non-negative numbers, or $\emptyset$ for the empty set) would make h_3 admissible?

10. (0.25 points) What range of possible values for $h_3(B)$ would make h_3 consistent?

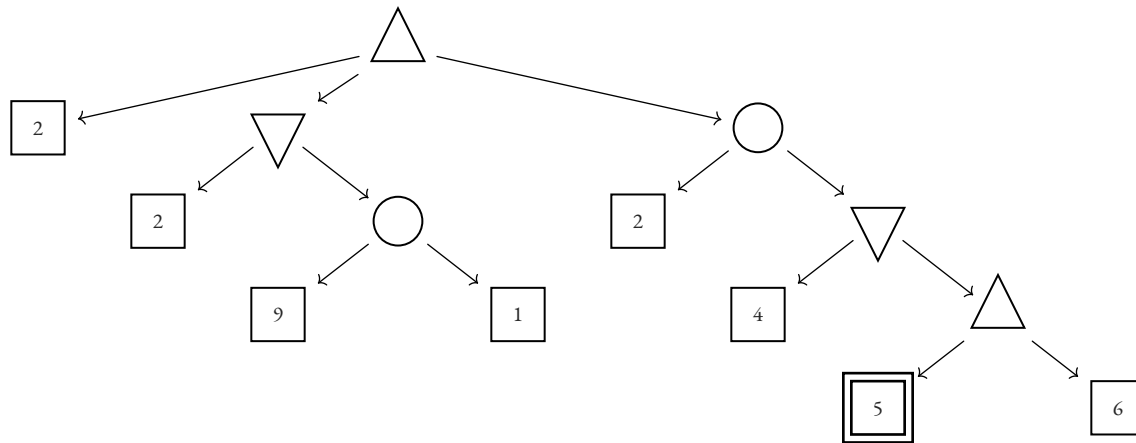
11. (0.25 points) What range of possible values for $h_3(B)$ would make the node expansion order of A^* graph search using heuristic h_3 start in order: A, C, B, D .

IV Game Search

12. (0.25 points) Draw the smallest possible game tree on which $\alpha\beta$ will prune at least one leaf node. Make sure to clearly state your search order, label the leaves with values, and mark the edges for the branch or branches that will be pruned.

13. (0.25 points) Imagine an agent is playing a game and choosing actions based on minimax. However, *unknownst to the agent*, their opponent has decided to play suboptimally. Would the actions chosen by the minimax agent in this scenario be better described as *optimistic* or *pessimistic*? Are there scenarios where the secretly suboptimal adversary will score better than an optimal adversary?

For the following questions consider a game with a complex mixture of maximizing, minimizing, and chance nodes (where all chance outcomes are equally likely). In this game, all possible utilities are values in the range $[0, 10]$, and each non-terminal node has at most three successors. When traversing the game tree below, use a left to right tie-breaking order.



14. (0.25 points) What is the game utility value at the root node of the above tree?
15. (0.25 points) Is pruning possible in this game? If not, briefly explain why. If so, cross out the branches that can be pruned.
16. (0.25 points) Whether or not pruning can occur, you can always still calculate values for α and β representing the best expected fallback scores for the maximizer and minimizer respectively at a given search node. What are the values for α and β after processing the highlighted node?